Root Cause Analysis (RCA) Document

GenAI AML Risk Detection System

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1. Overview

This document outlines key failure scenarios, root causes, impacts, and mitigation strategies identified in the GenAI-powered AML Risk Detection System. The goal is to ensure production readiness, robustness, and compliance alignment.

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2. System Context

The system consists of:

- Databricks Lakehouse (Bronze → Silver → Gold)
- FastAPI-based inference layer
- Multi-agent architecture (Retriever, Reasoner, Validator)
- RAG pipeline using FAISS
- LLMOps monitoring and logging
- Security layer (RBAC, masking, prompt guard)

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3. RCA Scenarios

3.1 Issue: No Relevant Context Retrieved

Observation: RAG pipeline returns empty or irrelevant policy documents.

Root Cause:

- Poor document chunking strategy
- Embedding mismatch between query and documents
- Low top-k retrieval value

Impact:

- Weak or incorrect reasoning
- Increased hallucination risk

Mitigation:

- Improve chunking with overlap

- Introduce metadata-based filtering
- Increase top-k retrieval
- Periodically refresh embeddings

3.2 Issue: Incorrect Risk Classification

Observation: Transactions classified into wrong risk categories.

Root Cause:

- Static rule-based scoring limitations
- Lack of contextual intelligence

Impact:

- Compliance violations
- False positives/negatives

Mitigation:

- Enhance validation agent logic
- Introduce ML-based scoring (future scope)
- Add threshold tuning

3.3 Issue: Hallucinated Reasoning Output

Observation: Generated explanations are generic or not grounded in policies.

Root Cause:

- Weak grounding in retrieved context
- Over-reliance on template reasoning

Impact:

- Reduced trust in system
- Poor auditability

Mitigation:

- Enforce context-only response generation
- Strengthen prompt templates
- Add response validation layer

3.4 Issue: High Latency

Observation: Slow response time during transaction analysis.

Root Cause:

- Sequential execution of agents
- No caching mechanism

Impact:

- Poor user experience
- Scalability limitations

Mitigation:

- Parallelize agent execution
- Cache embeddings and retrieval results
- Optimize model loading

3.5 Issue: Prompt Injection Attack

Observation: Malicious queries attempt to override system instructions.

Root Cause:

- Unvalidated user input

Impact:

- Data leakage risk
- System manipulation

Mitigation:

- Implement prompt injection guard
- Input sanitization
- Restrict system prompt exposure

3.6 Issue: Data Leakage Risk

Observation: Sensitive customer information exposed in outputs.

Root Cause:

- Missing data masking
- Improper access control

Impact:

- Regulatory violations
- Privacy breach

Mitigation:

- Apply masking on sensitive fields
- Enforce RBAC
- Audit logging for all access

3.7 Issue: FAISS Index Failure

Observation: Vector search fails or returns errors.

Root Cause:

- Index not loaded or corrupted
- Embedding dimension mismatch

Impact:

- RAG pipeline failure
- System downtime

Mitigation:

- Add fallback logic (default policies)
- Validate index at startup
- Monitor index health

3.8 Issue: API Failure / Downtime

Observation: FastAPI endpoint becomes unavailable.

Root Cause:

- Application crash
- Resource exhaustion

Impact:

- System unavailability

Mitigation:

- Add retry mechanisms
- Health check endpoints
- Logging and alerting

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4. Monitoring & Detection

- Logging via LLMOps tracker
- Latency tracking per agent
- Audit logs for all transactions
- Error logging for failures

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5. Preventive Measures

- Regular testing of RAG pipeline
- Validation of data pipelines
- Security testing (prompt injection scenarios)
- Performance benchmarking

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6. Conclusion

This RCA framework ensures that the GenAI AML system is resilient, secure, and aligned with enterprise production standards. Continuous monitoring and iterative improvements are essential for maintaining system reliability and compliance.